J-Anne Yow

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Summary

Final-year PhD candidate in Robotics & AI at NTU Singapore, developing adaptive robotic systems that integrate reinforcement learning, multimodal perception, and language-driven control. Experienced in sim-to-real transfer, ROS/ROS2, and real-world deployment in user trials. Passionate about bridging research and application by enabling embodied AI models to transfer from simulation into practical, deployable robots.

Education

Nanyang Technological University, Singapore — Sept 2021 – Feb 2026 (Expected)
Ph.D. in Mechanical Engineering

- **Supervisor:** Prof Ang Wei Tech
- **Dissertation:** Towards Personalized Robot Assistance: Integrating User Preferences in Robot-Assisted Feeding
- **Research Interests:** Human-Robot Interaction, Machine Learning, Foundation Models, Continual Learning

Nanyang Technological University, Singapore — Aug 2017 – May 2021
BE in Mechanical Engineering with a Second Major in Business

- **GPA:** 4.92/5.00, First Class Honours
- **Awards:** Dean's List every Academic Year, Dr Leung Shiu Kee Gold Medal Award, ASEAN Undergraduate Scholarship

Research Experience

Embodied AI Data Factory - Improving Teleoperation for Scalable Data Collection — 2025 – present

- Leading efforts to enhance teleoperation for bimanual and dexterous manipulation to reduce cognitive load and preserve natural human motion.
- Building a scalable pipeline with markerless motion capture, multisensory perception, and real-time retargeting to dexterous hands, manipulators, and humanoids in simulation and on real hardware.
- Goal: To close the gap between human and robot motion to enable efficient, high-fidelity datasets for training robust, generalizable, foundation model-based vision-language-action (VLA) policies.

Robot-Assisted Feeding — 2022 – present

- Led the technical delivery of an end-to-end adaptive feeding system, integrating perception, planning, and control into a safe, deployable platform for real-world elderly user trials.
- Developed goal-conditioned reinforcement learning policies in MuJoCo, integrating multimodal feedback (vision + force-torque sensing) for adaptive manipulation across variable food properties, achieving successful sim-to-real transfer.
- Built a language-driven adaptation pipeline, leveraging large language models to map user feedback into motion parameters for reliable, safe robot behaviour.

Grasping in Clutter — 2020 – 2023

- Developed a point-and-click interface for human-in-the-loop robotic manipulation, integrating object segmentation, grasp ranking, and grasp cycling to streamline user input and improve selection efficiency in cluttered scenes.
- Developed a POMDP-based shared autonomy framework that queries users under intent uncertainty, reducing conflicting actions and enabling faster task completion.

Work Experience

Research Assistant, Nanyang Technological University – Singapore — Aug 2021 – present

- Managed ROS/ROS2-based system integration and hardware-software interfacing, collaborating with research staff and students to define technical goals and evaluate prototypes in simulation and on physical robots.
- Managed project planning, resource coordination, and ethical compliance, removing operational bottlenecks such as compute limitations to keep development on schedule.

- Mentored and scoped eight undergraduate projects, defining research objectives, guiding technical execution and skill development.
- Drafted and contributed to competitive funding proposals to support ongoing research and new initiatives.

Selected Publications

SAVR: Scooping Adaptation for Variable food properties via Reinforcement Learning Oct 2025
J-Anne Yow, Wei Tech Ang
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

FRANC: Feeding Robot for Adaptive Needs and Personalized Care Oct 2025
J-Anne Yow, Luke Toh, Yi Heng San, Wei Tech Ang
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

ORBiT: Optimizing Robot-Assisted Bite Transfer Leveraging a Real2Sim2Real Framework Oct 2025
Sherwin Chan, ***J-Anne Yow***, Yi Heng San, Vasanth Ravichandram, Yifan Wang, Lek Syn Lim, Wei Tech Ang
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Shared Autonomy of a Robotic Manipulator for Grasping under Human Intent Uncertainty using POMDPs Nov 2023
J-Anne Yow, Neha P Garg, Wei Tech Ang
IEEE Transactions on Robotics (T-RO)

Service

Reviewer: IEEE Robotics and Automation Letters (RAL), IEEE International Conference on Robotics and Automation (ICRA)

Skills

Programming: Python, C/C++

Robotics & Simulation: ROS/ROS2, MoveIt, MuJoCo, Isaac Lab

Machine Learning: Reinforcement Learning, Imitation Learning, PyTorch

Perception & Control: Vision, Force-Torque Sensing, Impedance/Admittance Control

Languages: English, Chinese, Malay